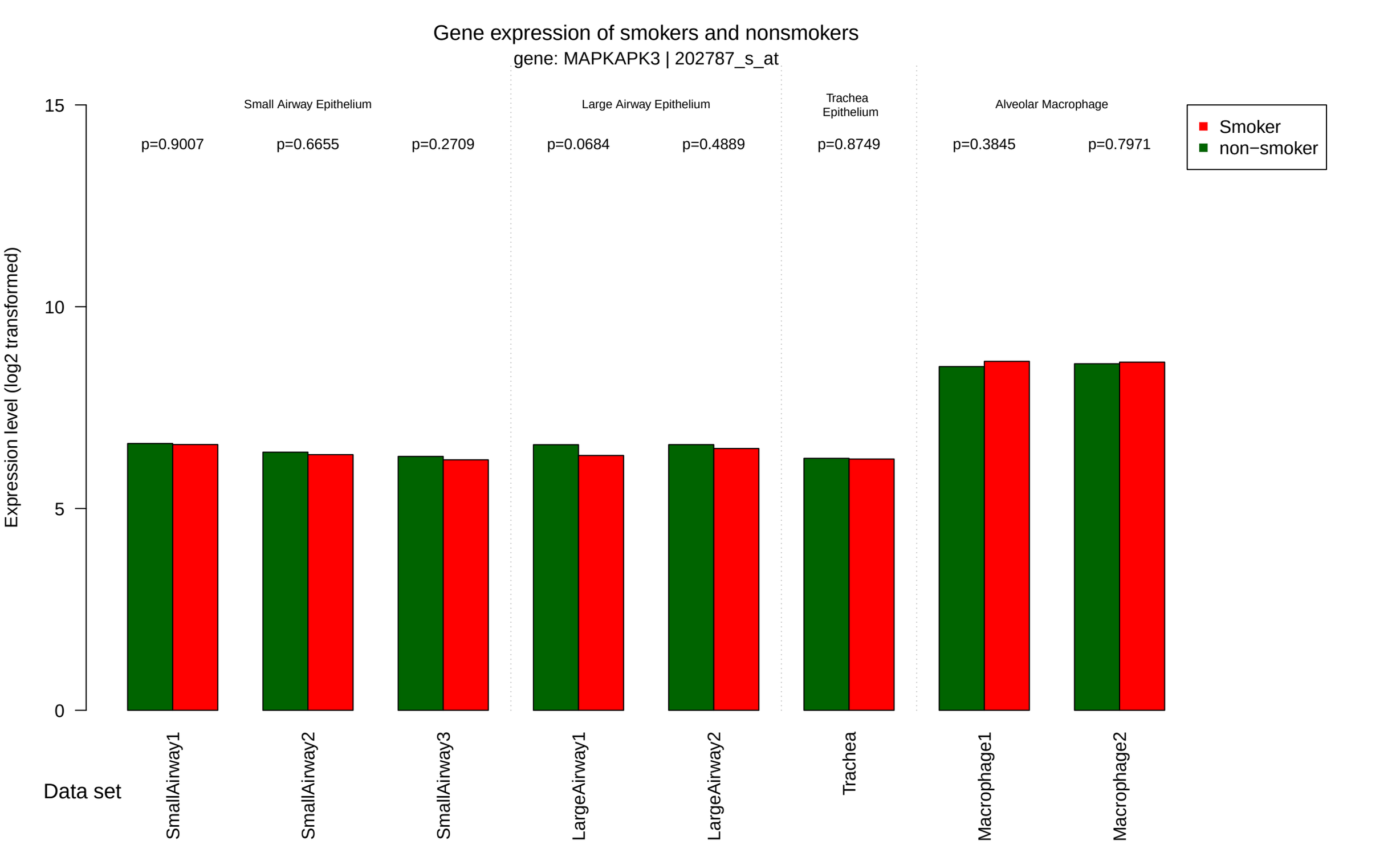
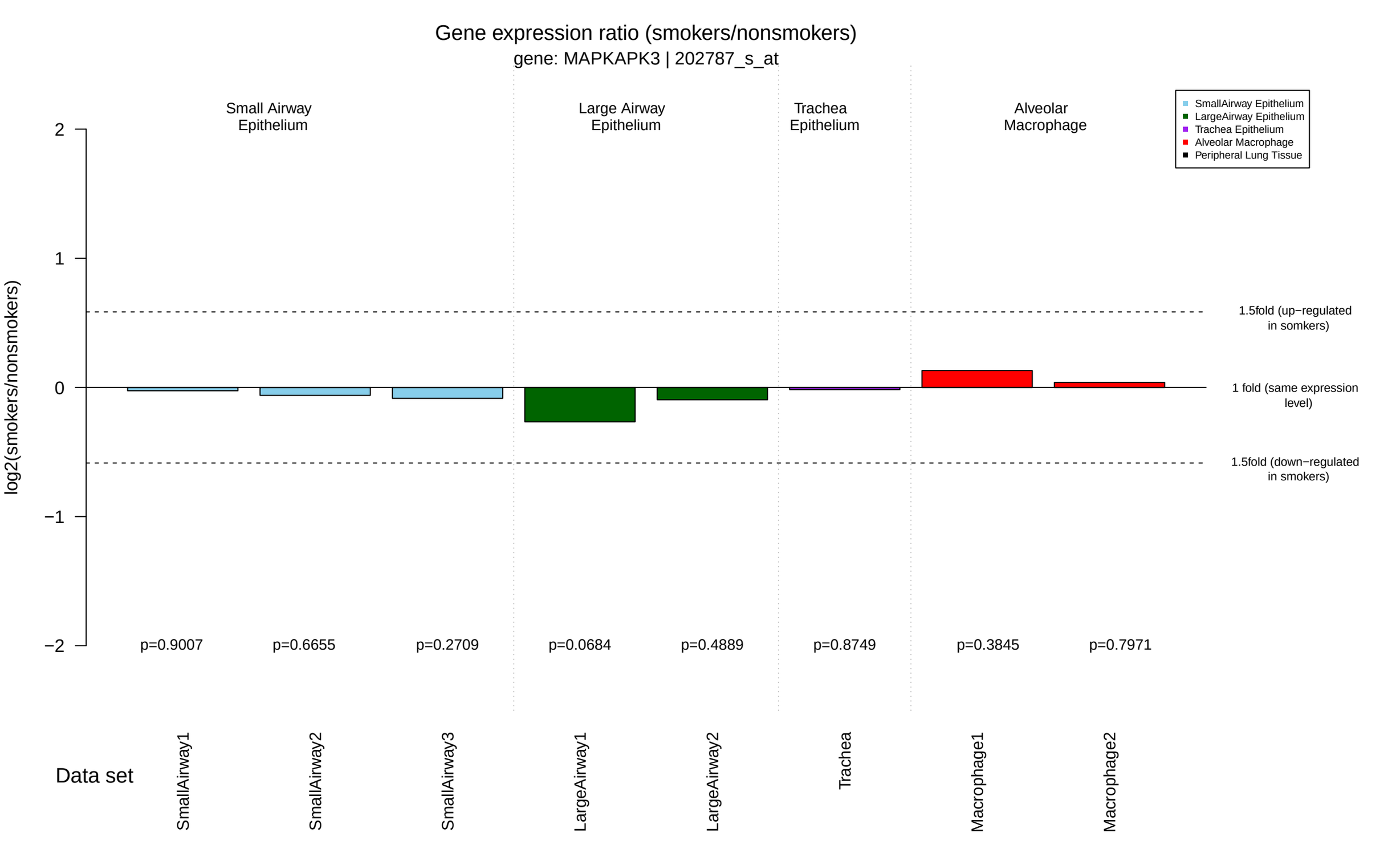
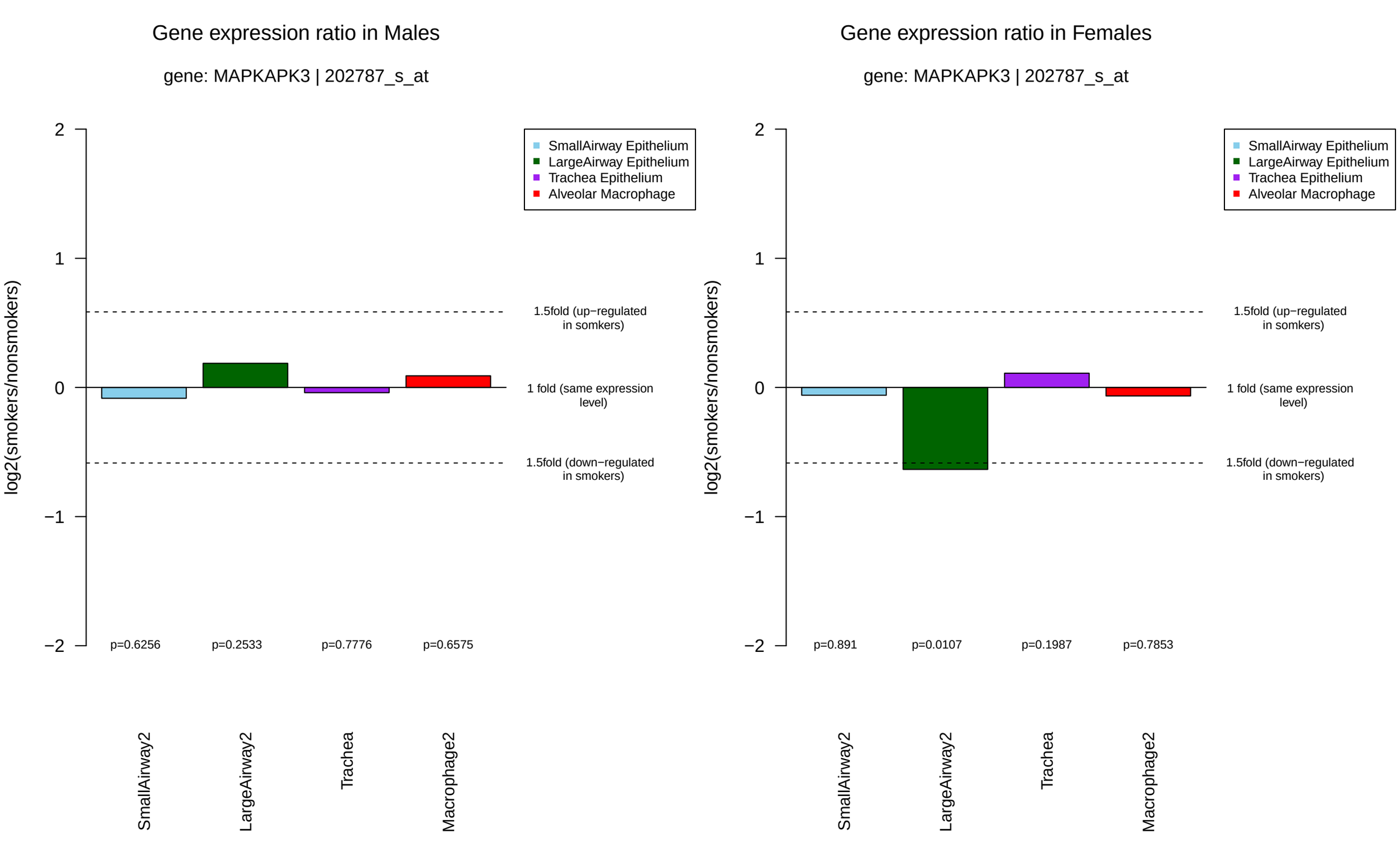
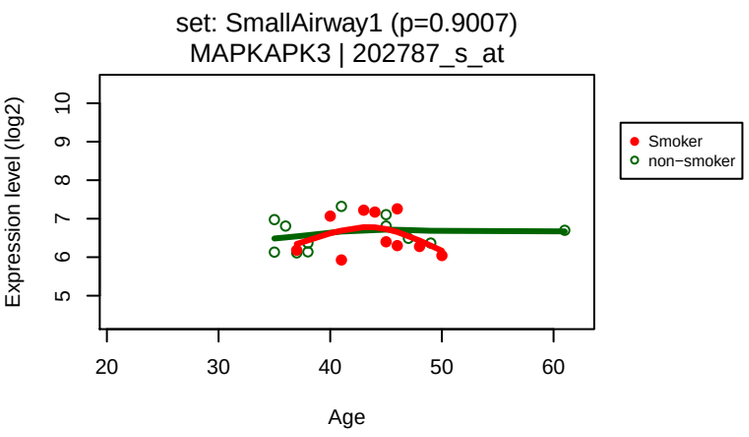
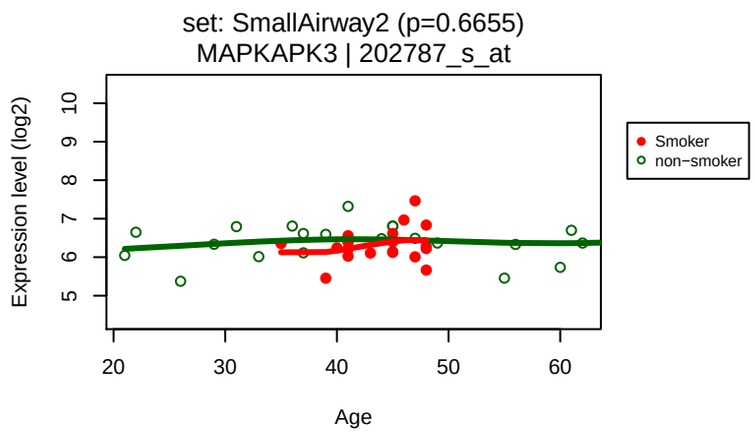
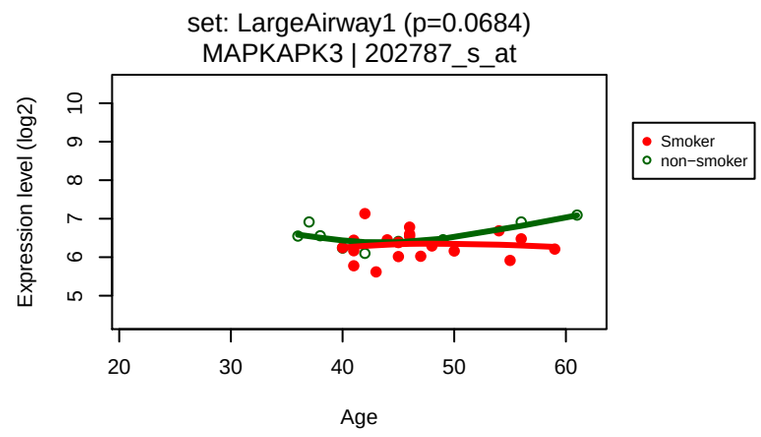

Gene expression of smokers and nonsmokers
gene: MAPKAPK3 | 202787_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9007
p=0.6655
p=0.2709
p=0.0684
p=0.4889
p=0.8749
p=0.3845
p=0.7971
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

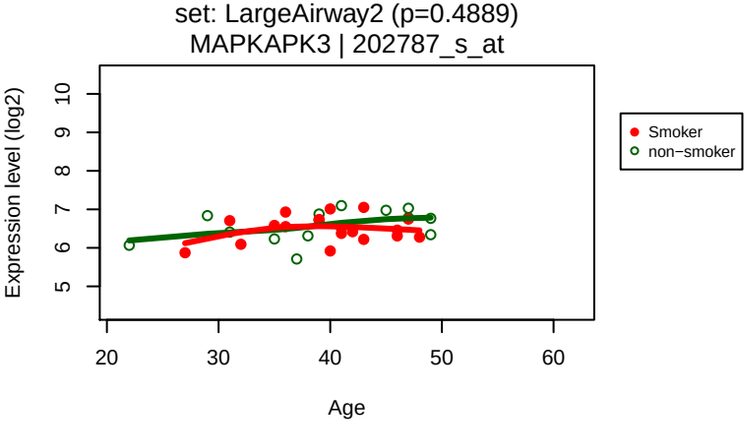

Gene expression ratio (smokers/nonsmokers)
gene: MAPKAPK3 | 202787_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9007
p=0.6655
p=0.2709
p=0.0684
p=0.4889
p=0.8749
p=0.3845
p=0.7971
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

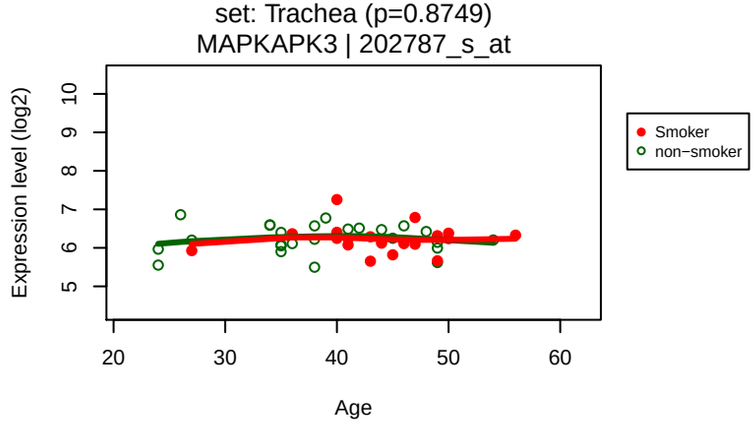

Gene expression ratio in Males
gene: MAPKAPK3 | 202787_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6256
p=0.2533
p=0.7776
p=0.6575
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPKAPK3 | 202787_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.891
p=0.0107
p=0.1987
p=0.7853
SmallAirway2
LargeAirway2
Trachea
Macrophage2

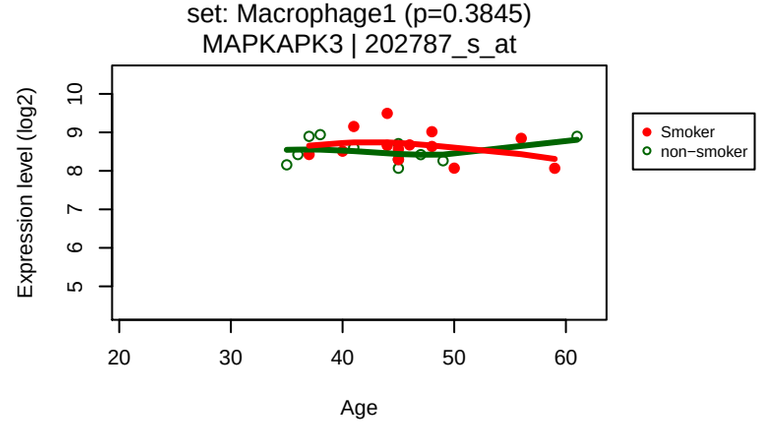

set: SmallAirway1 (p=0.9007)
MAPKAPK3 | 202787_s_at
Expression level (log2)
Age
Smoker
non−smoker

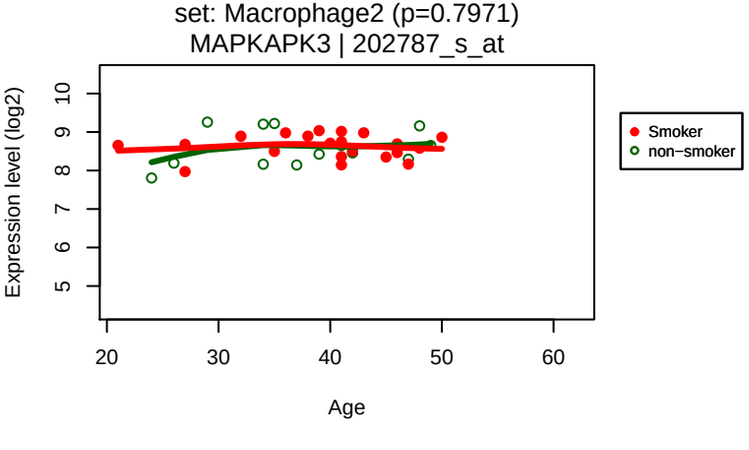

set: SmallAirway2 (p=0.6655)
MAPKAPK3 | 202787_s_at
Expression level (log2)
Age
Smoker
non−smoker

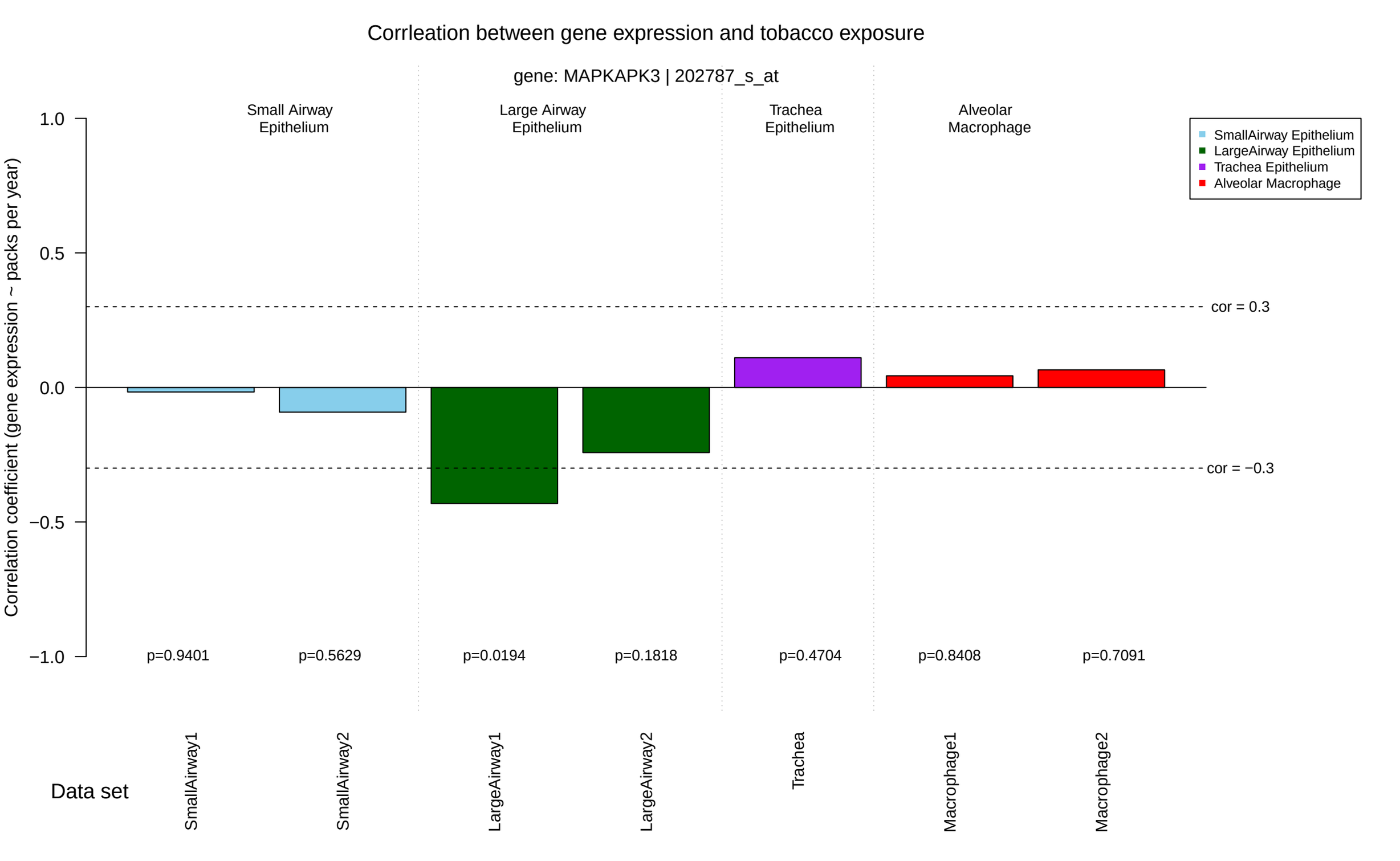

set: LargeAirway1 (p=0.0684)
MAPKAPK3 | 202787_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4889)
MAPKAPK3 | 202787_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8749)
MAPKAPK3 | 202787_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3845)
MAPKAPK3 | 202787_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7971)
MAPKAPK3 | 202787_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPKAPK3 | 202787_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9401
p=0.5629
p=0.0194
p=0.1818
p=0.4704
p=0.8408
p=0.7091
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2